

condition of water points in tanzania presentation



BUSINESS PRESENTATION



DATA UNDERSTANDING

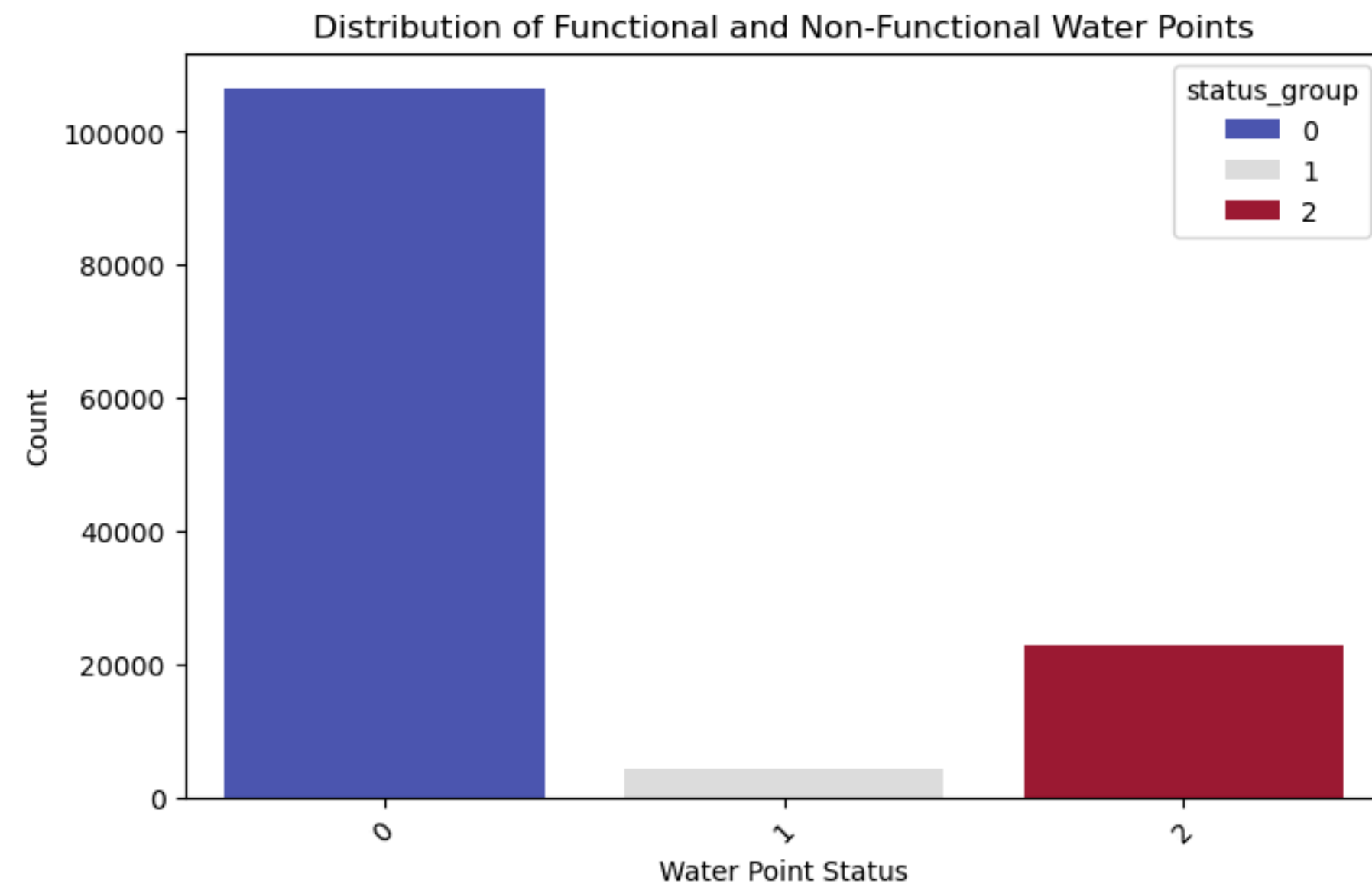
WATER POINT TYPE

1. WATER POINT TYPE
2. SOURCE
3. GEOGRAPHIC DISTRIBUTION
4. POPULATION AROUND THE WELL

INSTALLER

FUNCTIONALITY

```
plt.savefig("Distribution of Functional and Non-Functional Water Points")
```



<Figure size 640x480 with 0 Axes>



Search



9:29 PM
3/9/2025





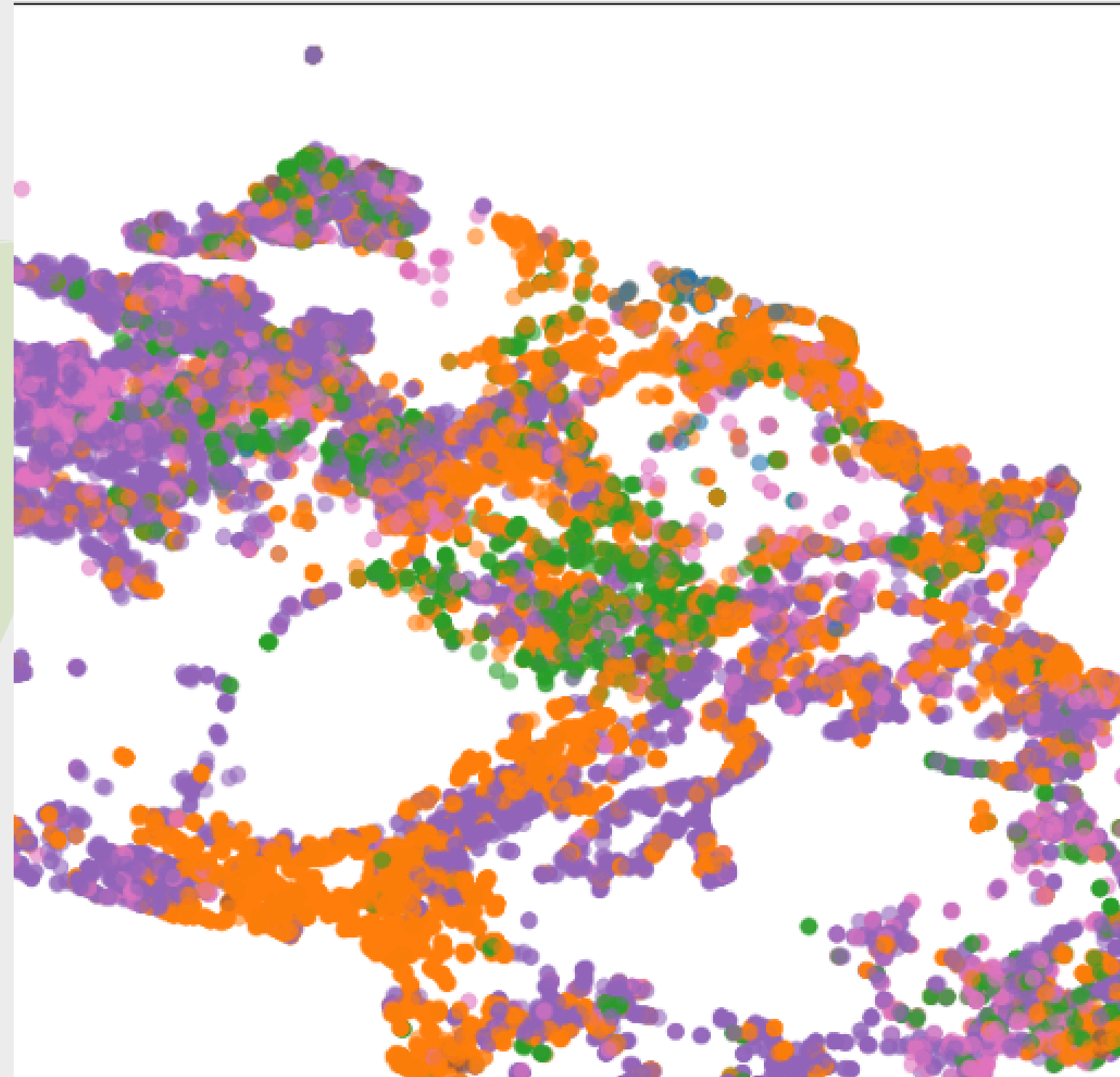
THIS DIAGRAM SHOWS
FUNCTIONALITY OF WATER
POINTS

Insight:
MOST POINTS ARE
FUNCTIONAL

ed.ipynb

utes ago

Geographic Distribution of Water Point Types



Search





**the scatterplot
shows geographic
distribution of
water point**

insight

water points are evenly
distributed

modeling



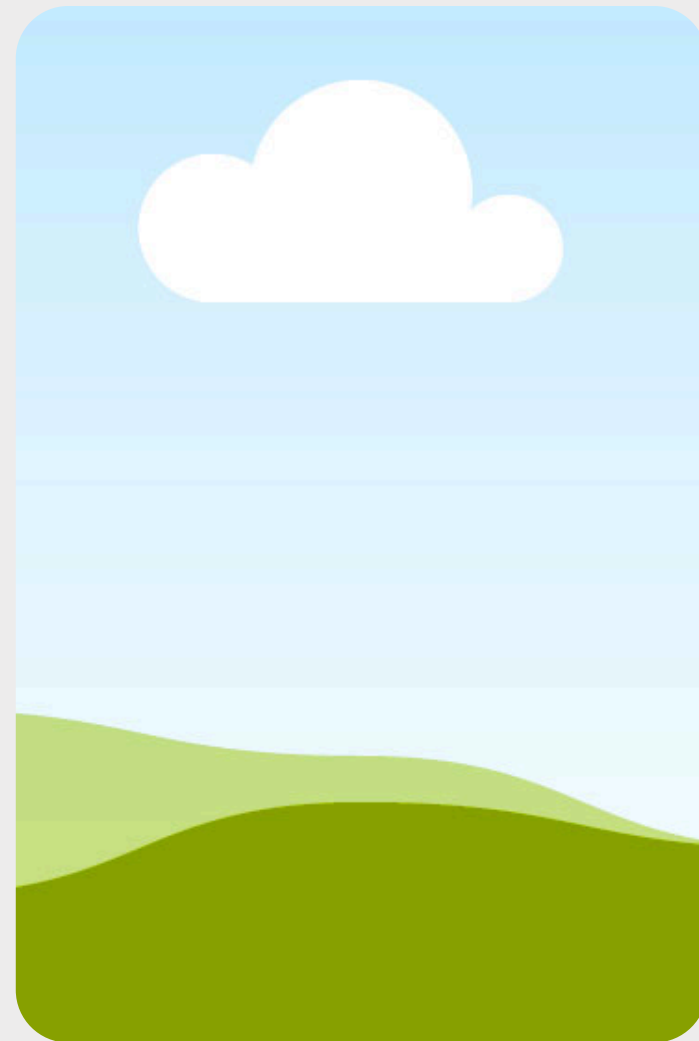
log regression
random forest
decision trees

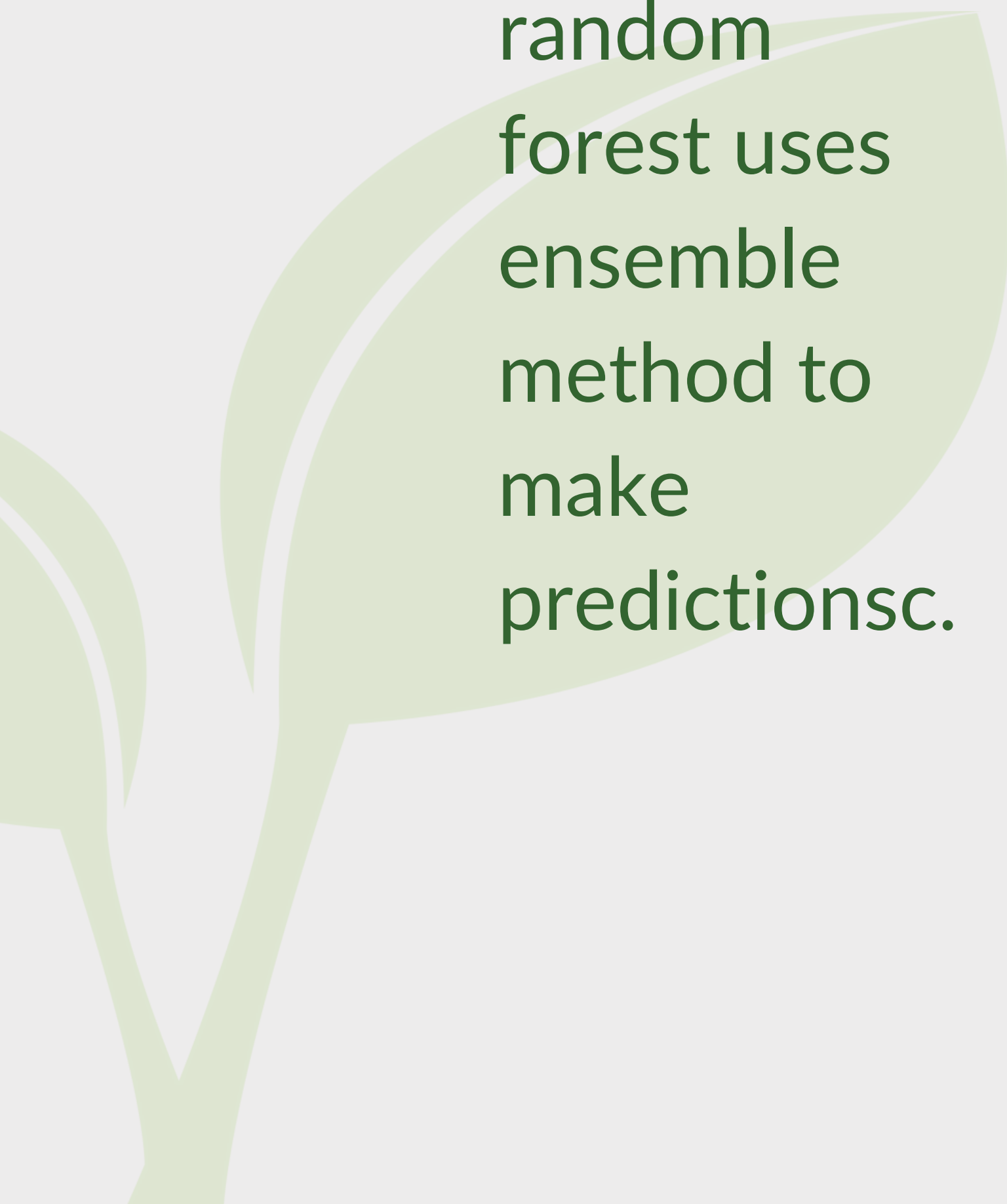
1lot

log regression

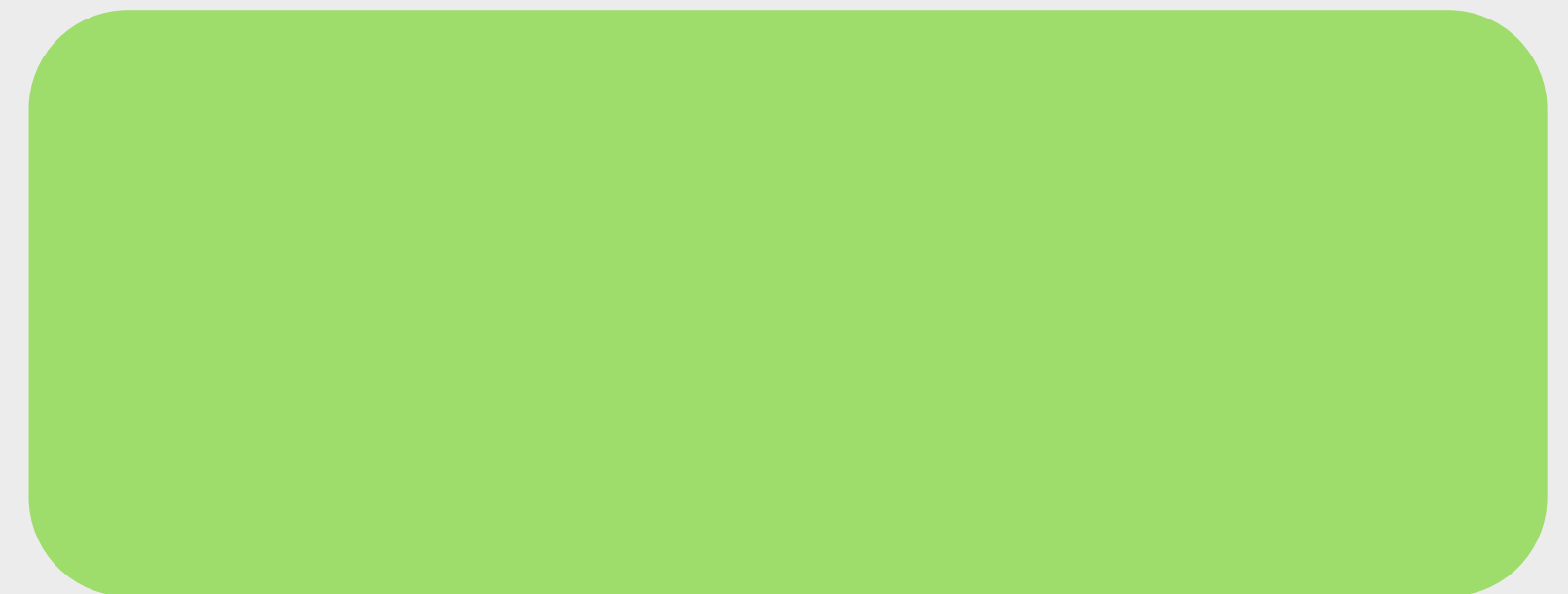
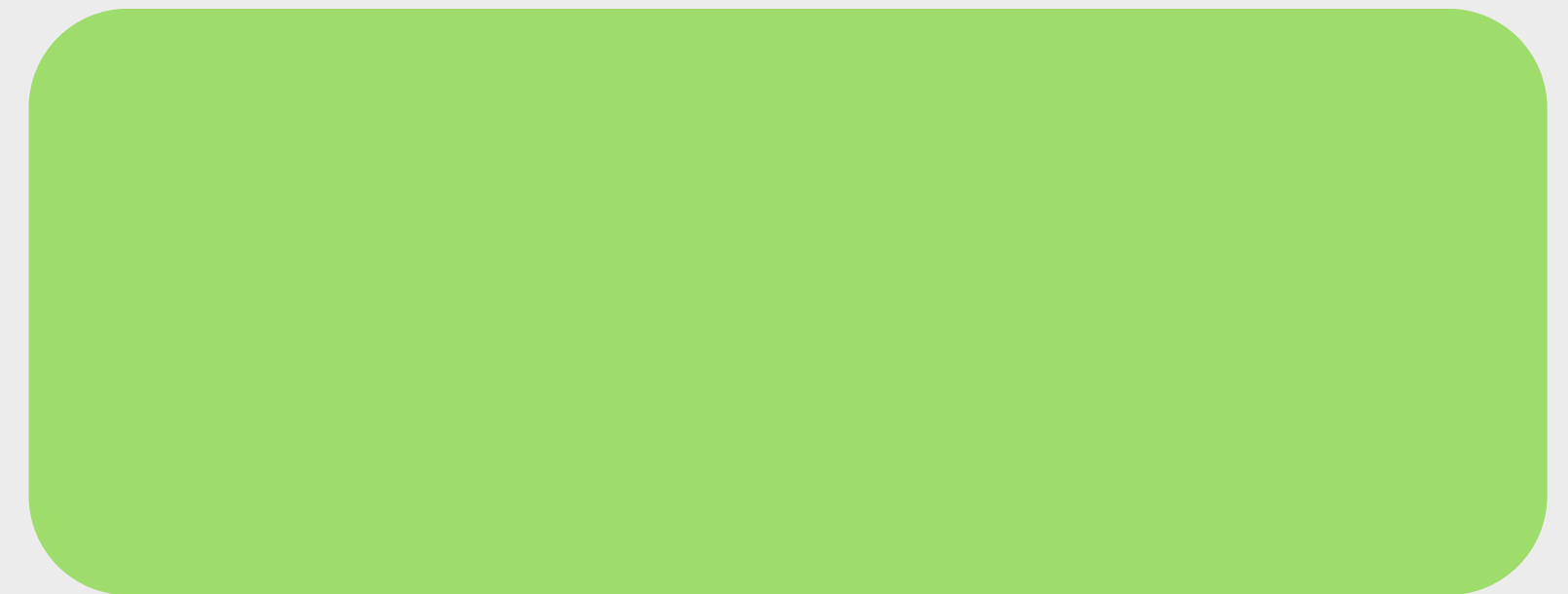


log regression is used to
predict a binary income ,a yes
or no



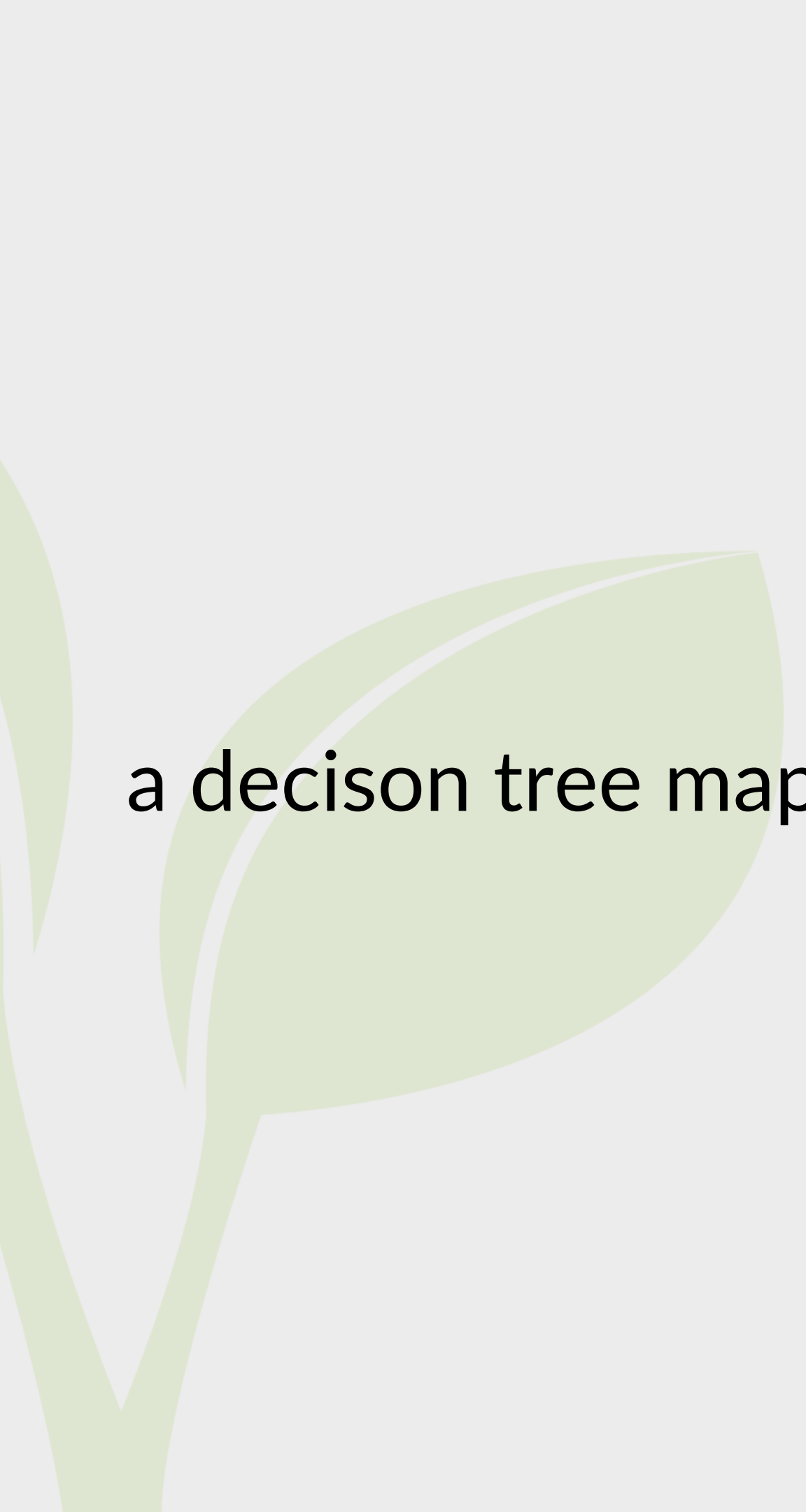
A decorative graphic on the left side of the slide, consisting of several overlapping, stylized leaves in shades of light green. The leaves are arranged in a vertical, branching pattern, with the largest leaf in the center and smaller ones branching off to the left and right.

random
forest uses
ensemble
method to
make
predictionsc.





decision trees



a decision tree maps out all possibilities out of a series of decision

Step 5

Elaborate on
the step here

MODEL PERFORMANCE TERMS

Precision: How many selected items are relevant?

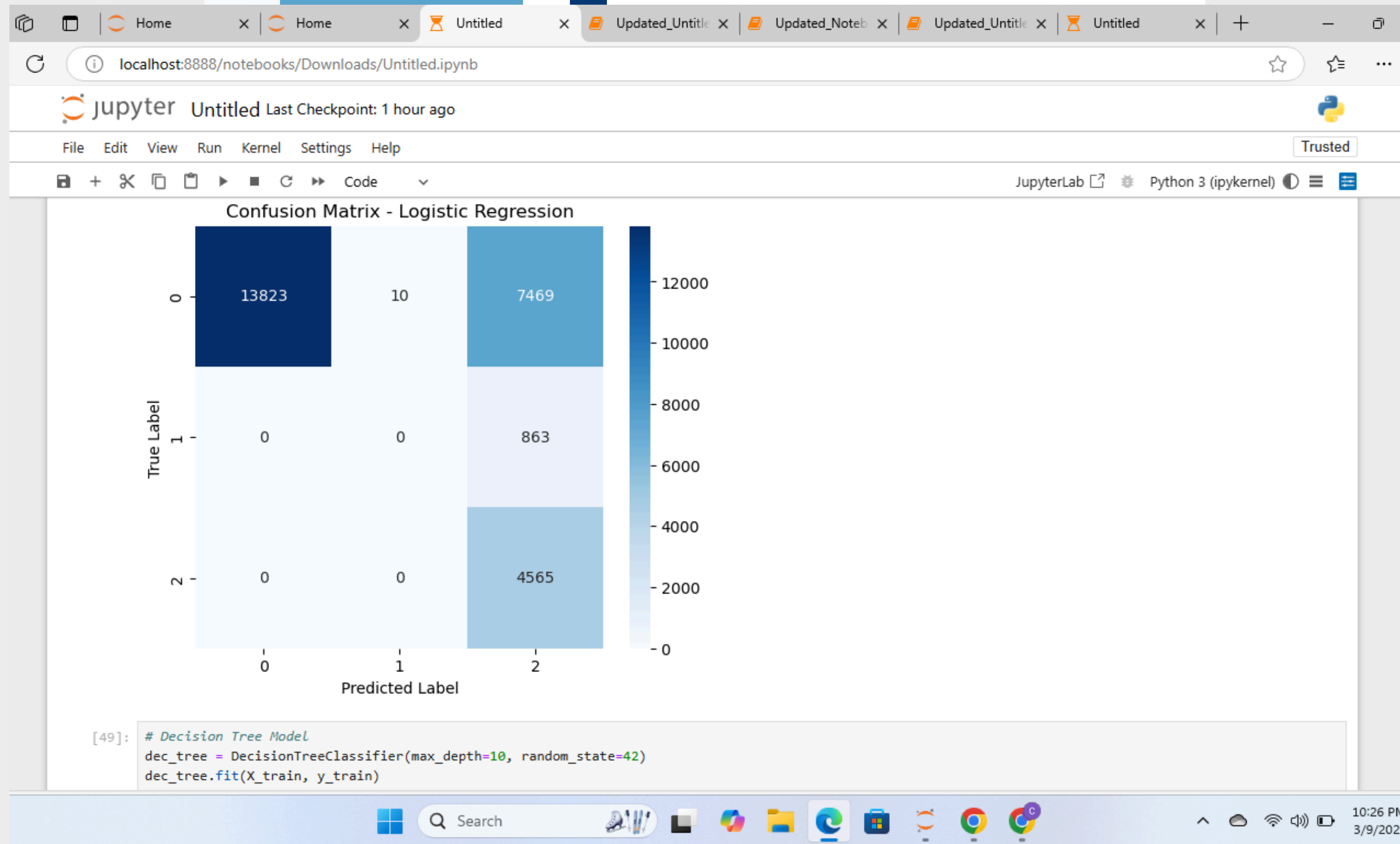
- **Recall:** How many relevant items were selected?
- **Accuracy:** Overall correctness of the model.

precision recall
f1-score support

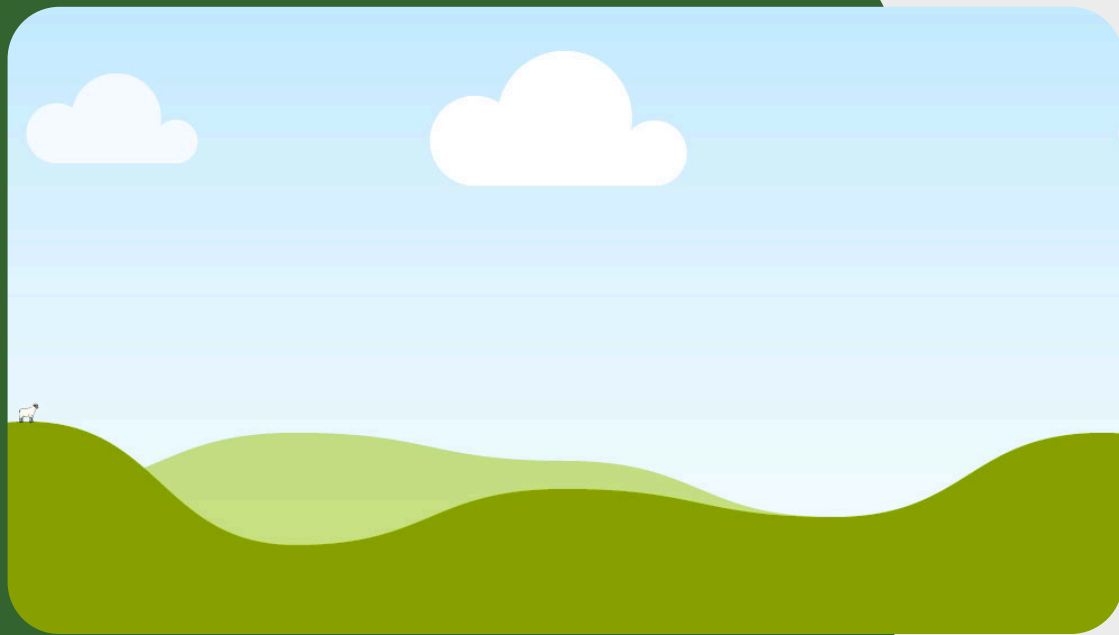
	0	1.00
0.65	0.79	
21302		

03.

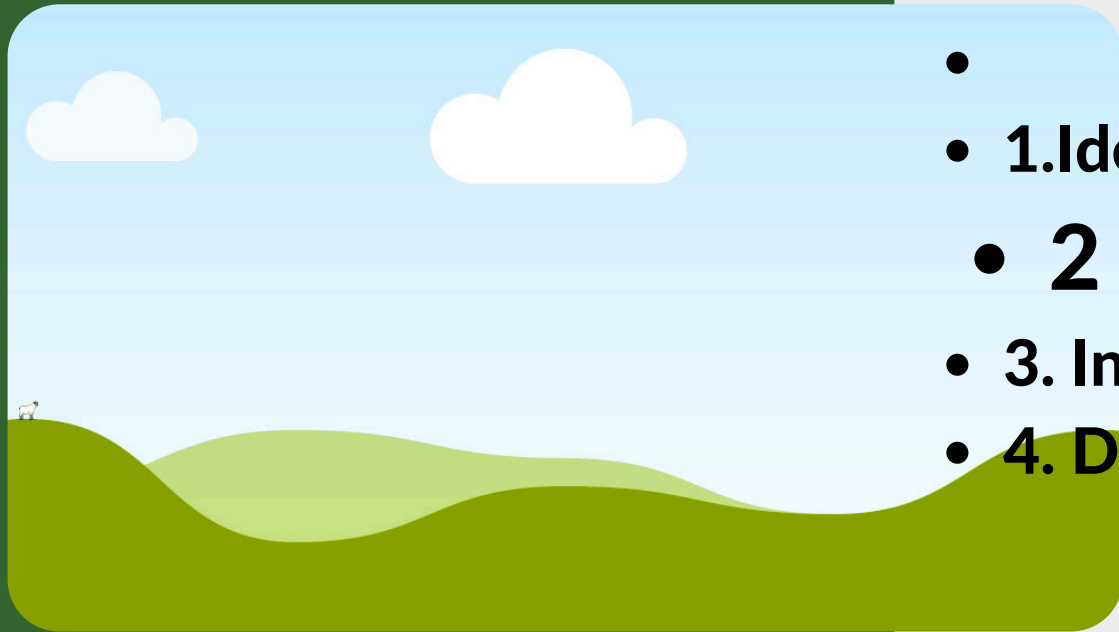
Logistic Regression



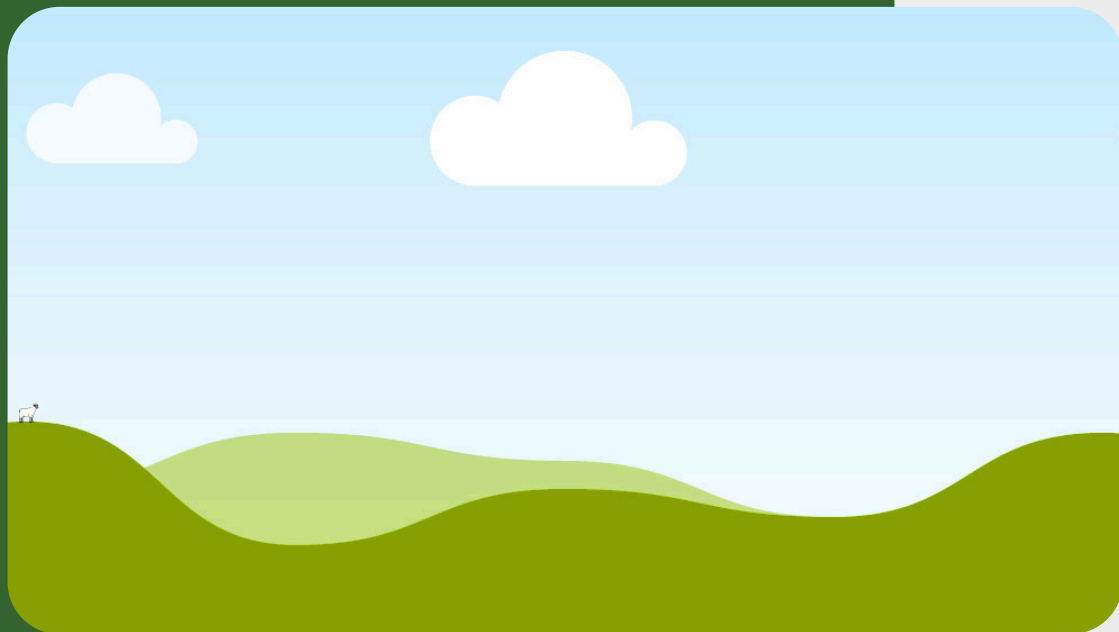
`ix_depth=10, random_state=42)`



- **RECOMMENDATIONS**



-
- 1. Identify key features highlighted by Random Forest as the basis for targeted interventions.
- **2 Enhance data gathering to minimize missing values.**
- 3. Increase maintenance activities in risky water points.
- 4. Deploy models to predict failures and act proactively. t



NEXT STEPS.

. Retrain the model repeatedly with fresh data.

Collaborate with field teams to validate predictions.

4.

contact
:chepkwonyjoy5
@gmail.com



**thank you
for your
time**

